CSE7302- University Project-Review 3

PROJECT TITLE—"AI-Powered Multi-Scan Medical Imaging and Report Interpretation System."

Batch Number: CSE-115

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I. System Architecture Overview
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    A. Image Standardization
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III. Modular Source Code Design
IV. Live Implementation Demonstration
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Algorithm 1: Image Pre-Processing Pipeline

Purpose:

Standardize medical images before AI inference.

Steps:

1. Input scan (CT/MRI/X-ray)
2. Resize to 224×224
3. Apply Gaussian noise reduction
4. Perform histogram equalization
5. Normalize pixel values (0–1)

Pseudocode:

Read image

Resize

Apply Gaussian Filter

Enhance Contrast

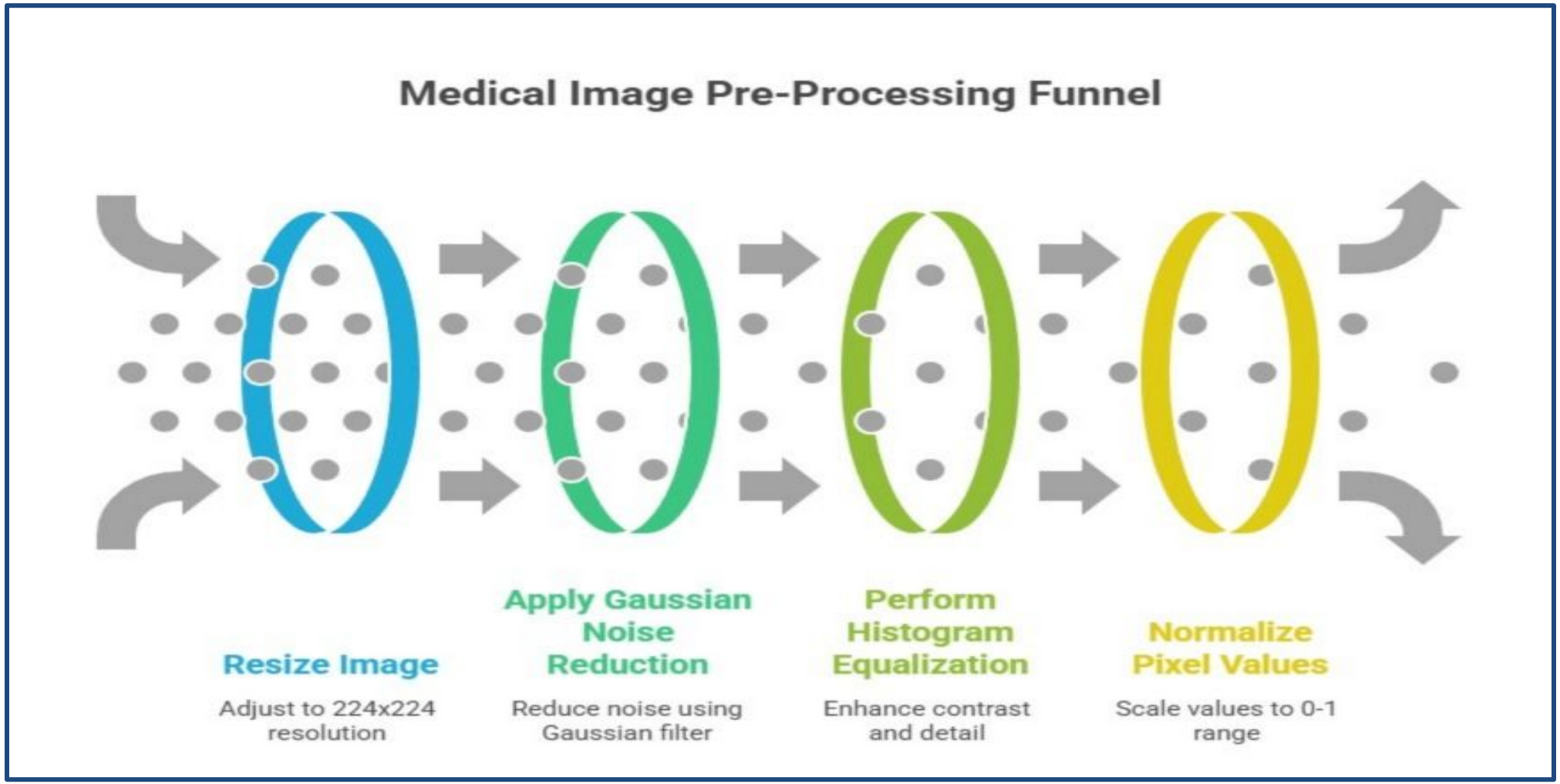
Normalize

Return Processed Image

Optimization:

- Reduces model bias
- Improves detection consistency
- Faster inference stability

Algorithm 1: Image Pre-Processing Pipeline



Algorithm 2: CNN-Based Diagnosis Model

Model:

Transfer Learning (ResNet /EfficientNet)

Flow:

Image →

Convolution Layers →
Feature Maps →
Dense Layer →
Softmax Output

Pseudocode:

Load Pretrained Model

Input Processed Image

Extract Features

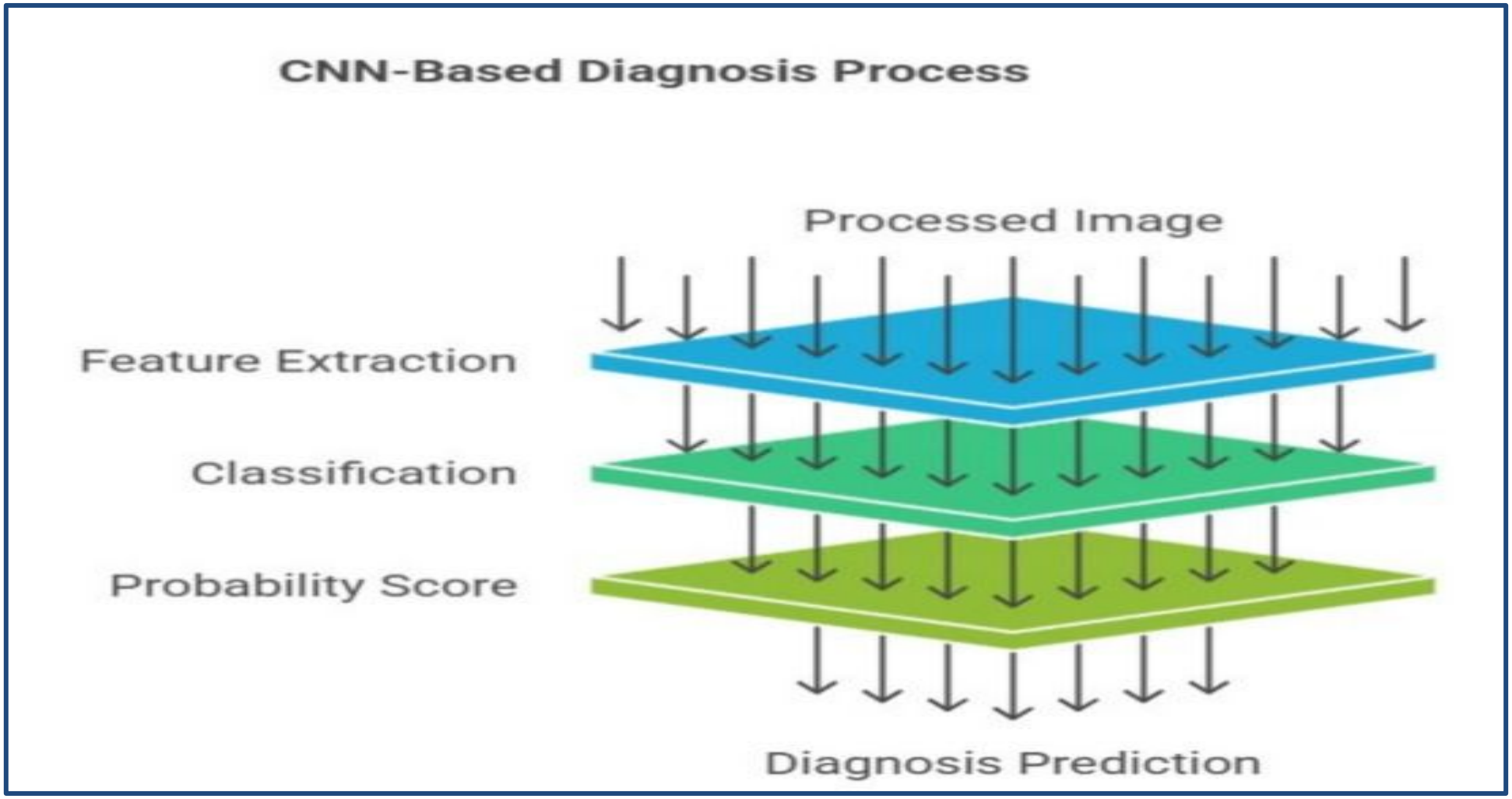
Apply Classification Layer

Return Probability Score

Optimization:

- Pretrained weights reduce training time
- Dropout prevents overfitting
- Confidence threshold filtering

Algorithm 2: CNN-Based Diagnosis Model



Algorithm 3: NLP Report Extraction

Purpose:

Extract structured medical insights from unstructured reports.

Steps:

1. Clean text
2. Tokenize
3. Extract medical entities
4. Identify lab values
5. Structure output

Pseudocode:

Input Report

Clean Text

Tokenize

Extract Entities (NER)

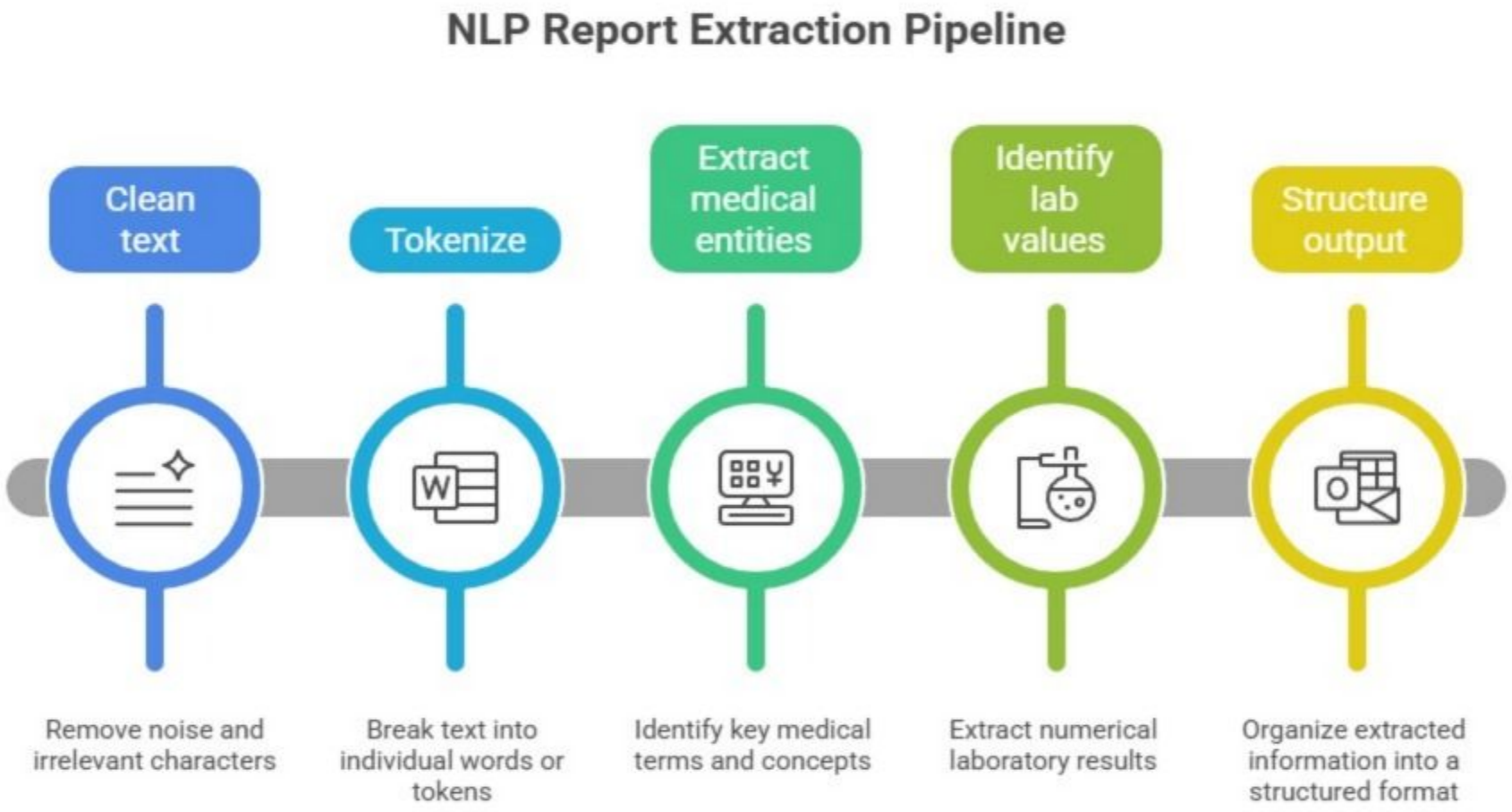
Extract Lab Values

Return Structured Report

Optimization:

- Zero-shot entity extraction
- Structured schema validation
- Error-handling for missing values

Algorithm 3: NLP Report Extraction



Algorithm 4: Multimodal Correlation Engine

Purpose:

Correlate image findings with lab biomarkers.

Logic:

For each anomaly:

- Identify related biomarker
- Compare against normal range
- Flag mismatch or confirm consistency

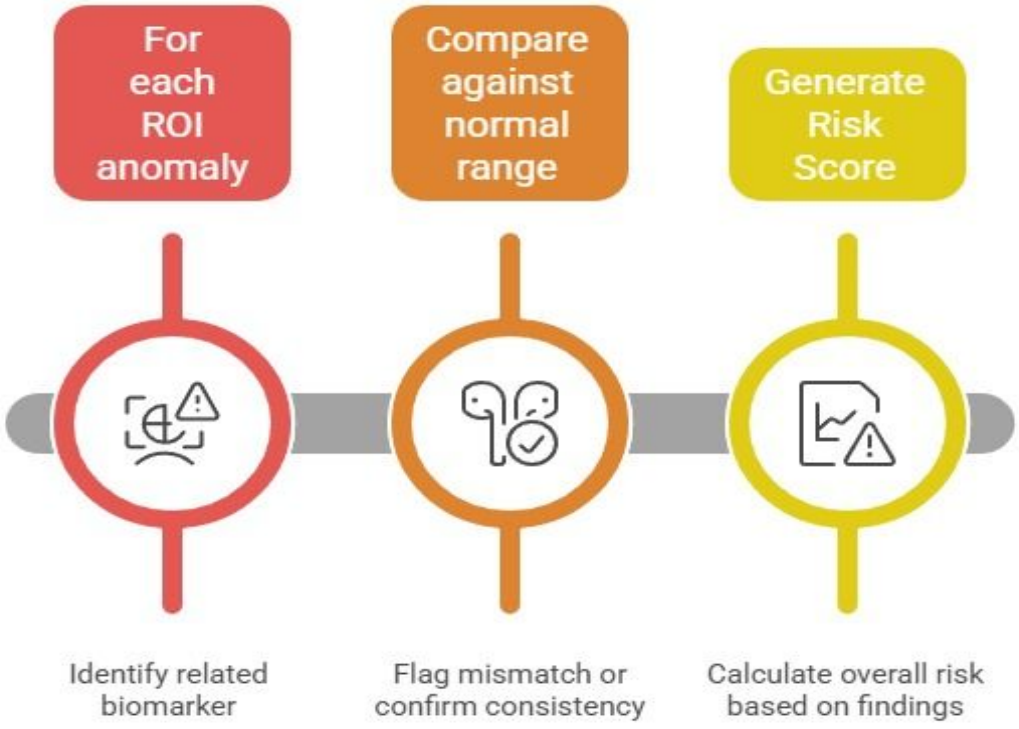
Pseudocode:

```
For each ROI anomaly:
  Fetch related biomarker
  If value outside range:
    Flag Inconsistency
  Else:
    Confirm Finding
Generate Risk Score
```

Optimization:

- Structured reasoning flow
- Controlled inference budget
- Deterministic fallback rules

Algorithm 4: Multimodal Correlation Engine



Algorithm 5: Transplant Matching & CIT Logic

Purpose:

Automate organ compatibility and Cold Ischemia Time risk.

Steps:

1. Calculate urgency score (MELD / LAS)
2. Check ABO compatibility
3. Check size compatibility
4. Compute estimated CIT
5. Rank by HLA match

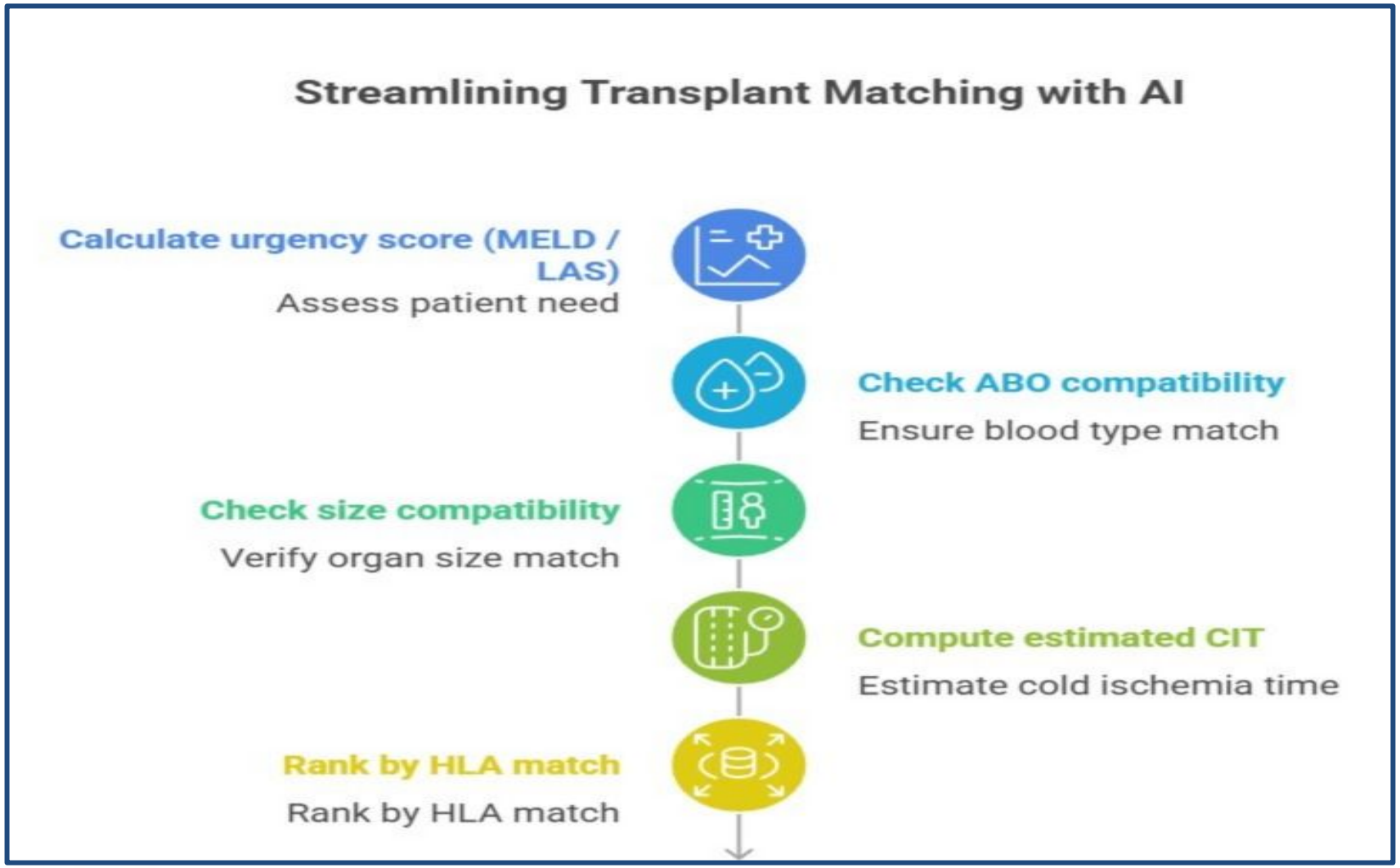
Pseudocode:

```
Compute Urgency Score
Check ABO Match
Check Size Compatibility
If Estimated_CIT > Threshold:
    Raise Critical Alert
Rank by HLA Score
Return Final Ranking
```

Optimization:

- Threshold-based alerting
- Multi-criteria ranking logic
- Compliance-aware validation

Algorithm 5: Transplant Matching & CIT Logic



Code Architecture

Folder Structure:

/services

/components

/models

/utils

/types

Engineering Principles

- Modular design
- Separation of concerns
- Typed schemas
- Well-commented functions
- Async error handling

Sample Code Snippet

```python
def analyze_scan(image):
    processed = preprocess(image)
    prediction = model.predict(processed)
    confidence = max(prediction)
    return confidence
```

Implemented Modules

Module	Status
Scan Upload	Done
Image Processing	Done
AI Diagnosis	Done
Report Parsing	Ongoing
Correlation Engine	Done
Dashboard	Done

Live Demo Workflow

1. Upload scan
2. System preprocesses image
3. CNN generates prediction
4. Upload lab report
5. Correlation engine analyzes
6. Risk score displayed

Output Includes:

- Confidence %
- Highlighted abnormality
- Structured summary

Draft Report Coverage

Includes:

- Abstract
- Literature Review (10 IEEE papers)
- Architecture
- All algorithms (detailed)
- Implementation screenshots
- Performance metrics

Conclusion

- Algorithms clearly defined
- Optimized and modular
- 50%+ working system
- Live demo ready
- Draft report submitted

Thank
You!